

S5: Hypothesis Testing — Binomial

1. Purpose of Hypothesis Testing

A **hypothesis test** uses sample data to assess whether there is sufficient statistical evidence to challenge a claim about a population parameter — in this case, the proportion p .

Null hypothesis H_0 : The default assumption; states the value of p if nothing has changed.

Example: $H_0 : p = 0.3$

Alternative hypothesis H_1 : What the researcher believes may be true; determines whether the test is one- or two-tailed.

2. One-Tailed vs Two-Tailed Tests

Test type	Alternative hypothesis	Significance level used
One-tailed (upper)	$H_1 : p > k$	Full α in upper tail
One-tailed (lower)	$H_1 : p < k$	Full α in lower tail
Two-tailed	$H_1 : p \neq k$	$\alpha/2$ in each tail

Exam Tip: The wording of the question tells you which tail to use. "Increased", "more likely", "greater" → upper tail. "Decreased", "less likely", "fewer" → lower tail. "Changed", "different" → two-tailed (use $\alpha/2$).

3. Significance Level

The **significance level** α (typically 5% or 1%) is the maximum probability of incorrectly rejecting H_0 when it is true (Type I error). We reject H_0 if the p-value $< \alpha$.

4. The Test Statistic

Under H_0 , assume $X \sim B(n, p_0)$ where p_0 is the value stated in H_0 . The test statistic is the **observed value** of X from the sample.

5. Two Methods to Carry Out the Test

Method 1: p-value Method

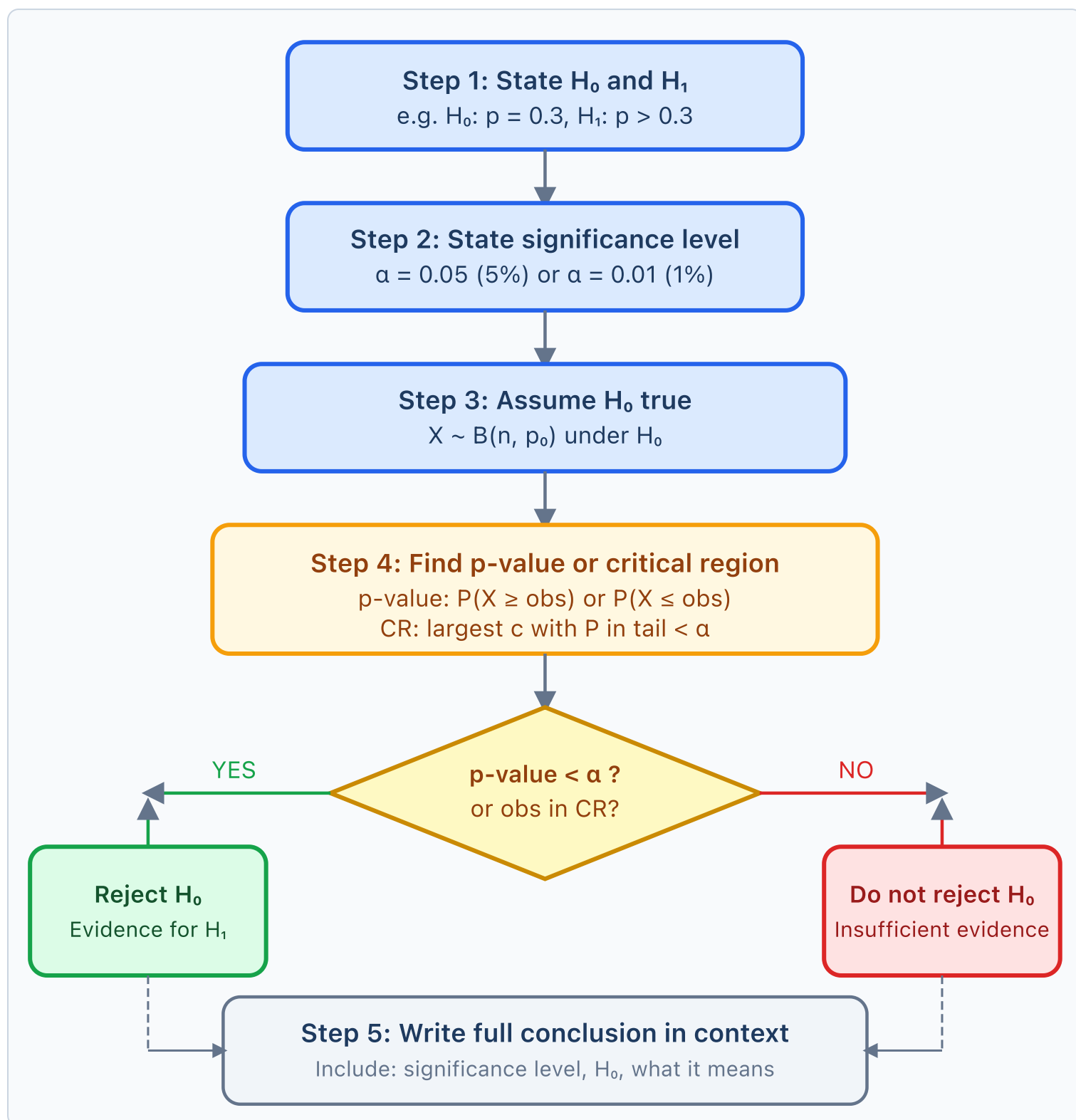
1. Find the probability of obtaining a result at least as extreme as the observed value, assuming H_0 is true.
2. For a lower tail test: p-value = $P(X \leq x_{\text{obs}})$
3. For an upper tail test: p-value = $P(X \geq x_{\text{obs}})$
4. If p-value $< \alpha$: **reject** H_0 .
5. If p-value $\geq \alpha$: **do not reject** H_0 .

Method 2: Critical Region Method

1. Find the **critical value** c such that $P(X \leq c) < \alpha$ (lower tail) or $P(X \geq c) < \alpha$ (upper tail).
2. The critical region is all values of X that lead to rejection of H_0 .
3. If observed value falls in the critical region: **reject** H_0 .

Exam Tip: The critical region is the largest set of values satisfying the probability condition. Because the binomial is discrete, the actual significance level (probability of the critical region under H_0) will usually be less than the nominal α .

6. Hypothesis Test Flowchart



7. Full Worked Example

Question: A coin is suspected to be biased in favour of heads. In 20 flips, it lands on heads 15 times. Test at the 5% significance level whether there is evidence that $p > 0.5$.

Step 1: $H_0 : p = 0.5$, $H_1 : p > 0.5$ (one-tailed, upper)

Step 2: Significance level $\alpha = 0.05$

Step 3: Under H_0 : $X \sim B(20, 0.5)$. Observed value $x = 15$.

Step 4 (p-value method):

$$p\text{-value} = P(X \geq 15) = 1 - P(X \leq 14)$$

Using calculator: $P(X \leq 14) = 0.9793$

$$p\text{-value} = 1 - 0.9793 = 0.0207$$

Decision: $0.0207 < 0.05$, so we reject H_0 .

Conclusion: "There is sufficient evidence at the 5% significance level to reject H_0 and conclude that the coin is biased towards heads (i.e., $p > 0.5$)."

8. Critical Region Example

Question: Using the same scenario ($X \sim B(20, 0.5)$, upper-tail test at 5%), find the critical region.

We need the smallest c such that $P(X \geq c) < 0.05$:

$$P(X \geq 14) = 1 - P(X \leq 13) = 1 - 0.9423 = 0.0577 > 0.05 \text{ — not in CR}$$

$$P(X \geq 15) = 0.0207 < 0.05 \text{ — in CR}$$

Critical region: $X \geq 15$

Actual significance level: $P(X \geq 15) = 0.0207$ (2.07%)

Since $x_{\text{obs}} = 15$ is in the CR, we reject H_0 . Same conclusion as above.

9. Writing the Conclusion — Template

Rejecting H_0 : "There is sufficient evidence at the $[x]\%$ significance level to reject H_0 and to suggest that $[interpret H_1 in context]."$

Not rejecting H_0 : "There is insufficient evidence at the $[x]\%$ significance level to reject H_0 . There is no significant evidence that $[interpret H_1 in context]."$

Exam Tip: Never say "accept H_0 " — always say "do not reject H_0 " or "there is insufficient evidence to reject H_0 ". You cannot prove a null hypothesis true.

10. Type I and Type II Errors

Error Type	Definition	Probability
Type I Error	Rejecting H_0 when it is actually true (false positive)	= actual significance level = $P(\text{CR} H_0 \text{ true})$
Type II Error	Failing to reject H_0 when it is actually false (false negative)	Depends on true value of p

Type I error example: From the worked example, if the coin is actually fair ($p = 0.5$) but we observe $X = 15$, the probability of incorrectly rejecting $H_0 = 0.0207$ (the actual significance level).

Exam Tip: The probability of a Type I error equals the actual significance level (probability that the observed value lies in the critical region given H_0 is true). This is NOT always equal to the nominal α for discrete distributions.

11. Two-Tailed Test Example

Question: A manufacturer claims a machine produces 20% defective items. A sample of 25 items is taken and 9 are defective. Test at the 5% level whether there is evidence the proportion has changed.

$H_0 : p = 0.2, H_1 : p \neq 0.2$ (two-tailed). $\alpha/2 = 0.025$ in each tail.

Under $H_0: X \sim B(25, 0.2)$. Observed $x = 9$ (upper tail, since $9 > np = 5$).

$$P(X \geq 9) = 1 - P(X \leq 8) = 1 - 0.9532 = 0.0468$$

Compare with $\alpha/2 = 0.025$: $0.0468 > 0.025$

Do not reject H_0 . "There is insufficient evidence at the 5% significance level to suggest the proportion of defective items has changed."